

48—(MathAA/20_HL_Winter_2022_Q11)—Algebra-functions-Roots,Differentiation

[Maximum mark: 15]

The function f is defined by $f(x) = e^{-x}(3x + 4)$, where $x \in \mathbb{R}$.

(a) Find $f'(x)$. [3]

(b) Hence or otherwise, find the coordinates of the point on the graph of $y = f(x)$ where the tangent is parallel to the line $y = x$. [3]

The region enclosed by the curve $y = f(x)$, the x -axis and the y -axis is rotated through 2π radians about the x -axis to form a solid of revolution.

(c) Find the volume of this solid. [4]

Consider a function g such that $g(0) = 1$ and $g'(0) = 2$.

(d) Find the value of

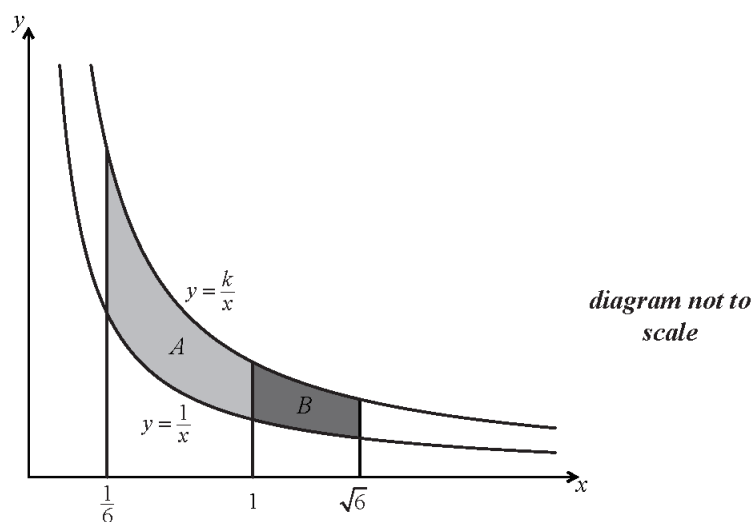
(i) $(f + g)(0)$;

(ii) $(f + g)'(0)$. [5]



1-(MAT/11_HL_Summer_2012_Q6)-Integration

The graph below shows the two curves $y = \frac{1}{x}$ and $y = \frac{k}{x}$, where $k > 1$.



- (a) Find the area of region A in terms of k . [3 marks]
- (b) Find the area of region B in terms of k . [2 marks]
- (c) Find the ratio of the area of region A to the area of region B . [3 marks]

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2-(MAT/12_HL_Summer_2012_Q10)-Functions-Roots, Graphs, Integration

The function f is defined on the domain $\left[0, \frac{3\pi}{2}\right]$ by $f(x) = e^{-x} \cos x$.

(a) State the two zeros of f .

[1 mark]

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(b) Sketch the graph of f .

[1 mark]



- (c) The region bounded by the graph, the x -axis and the y -axis is denoted by A and the region bounded by the graph and the x -axis is denoted by B . Show that the ratio of the area of A to the area of B is

$$\frac{e^{\pi} \left(e^{\frac{\pi}{2}} + 1 \right)}{e^{\pi} + 1},$$

[7 marks]

[illegible]

3-(MAT/11_HL_Summer_2012_Q12)-Differentiation,Integration

Let $f(x) = \sqrt{\frac{x}{1-x}}$, $0 < x < 1$.

- (a) Show that $f'(x) = \frac{1}{2}x^{-\frac{1}{2}}(1-x)^{-\frac{3}{2}}$ and deduce that f is an increasing function. [5 marks]
- (b) Show that the curve $y = f(x)$ has one point of inflexion, and find its coordinates. [6 marks]
- (c) Use the substitution $x = \sin^2 \theta$ to show that $\int f(x) dx = \arcsin \sqrt{x} - \sqrt{x-x^2} + c$. [11 marks]



4-(MAT/12_HL_Summer_2013_Q1)-Integration

[Maximum mark: 6]

Find the exact value of $\int_1^2 \left((x-2)^2 + \frac{1}{x} + \sin \pi x \right) dx$.

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5-(MAT/11_HL_Summer_2013_Q10)-Integration

- (a) Find all values of x for $0.1 \leq x \leq 1$ such that $\sin(\pi x^{-1}) = 0$. [2 marks]
- (b) Find $\int_{\frac{1}{n+1}}^{\frac{1}{n}} \pi x^{-2} \sin(\pi x^{-1}) dx$, showing that it takes different integer values when n is even and when n is odd. [3 marks]
- (c) Evaluate $\int_{0.1}^1 \left| \pi x^{-2} \sin(\pi x^{-1}) \right| dx$. [2 marks]

[illegible]

6-(MAT/11_HL_Summer_2013_Q12)-Quadratics, Graphs, Integration

(a) Express $4x^2 - 4x + 5$ in the form $a(x - h)^2 + k$ where $a, h, k \in \mathbb{Q}$. [2 marks]

(b) The graph of $y = x^2$ is transformed onto the graph of $y = 4x^2 - 4x + 5$. Describe a sequence of transformations that does this, making the order of transformations clear. [3 marks]

The function f is defined by $f(x) = \frac{1}{4x^2 - 4x + 5}$.

(c) Sketch the graph of $y = f(x)$. [2 marks]

(d) Find the range of f . [2 marks]

(e) By using a suitable substitution show that $\int f(x) dx = \frac{1}{4} \int \frac{1}{u^2 + 1} du$. [3 marks]

(f) Prove that $\int_1^{3.5} \frac{1}{4x^2 - 4x + 5} dx = \frac{\pi}{16}$. [7 marks]



7-(MAT/10_HL_Winter_2013_Q12)-ComplexNumbers,Trigonometry,Integration

Consider the complex number $z = \cos \theta + i \sin \theta$.

(a) Use De Moivre's theorem to show that $z^n + z^{-n} = 2 \cos n\theta$, $n \in \mathbb{Z}^+$. [2]

(b) Expand $(z + z^{-1})^4$. [1]

(c) Hence show that $\cos^4 \theta = p \cos 4\theta + q \cos 2\theta + r$, where p , q and r are constants to be determined. [4]

(d) Show that $\cos^6 \theta = \frac{1}{32} \cos 6\theta + \frac{3}{16} \cos 4\theta + \frac{15}{32} \cos 2\theta + \frac{5}{16}$. [3]

(e) Hence find the value of $\int_0^{\frac{\pi}{2}} \cos^6 \theta \, d\theta$. [3]

The region S is bounded by the curve $y = \sin x \cos^2 x$ and the x -axis between $x = 0$ and $x = \frac{\pi}{2}$.

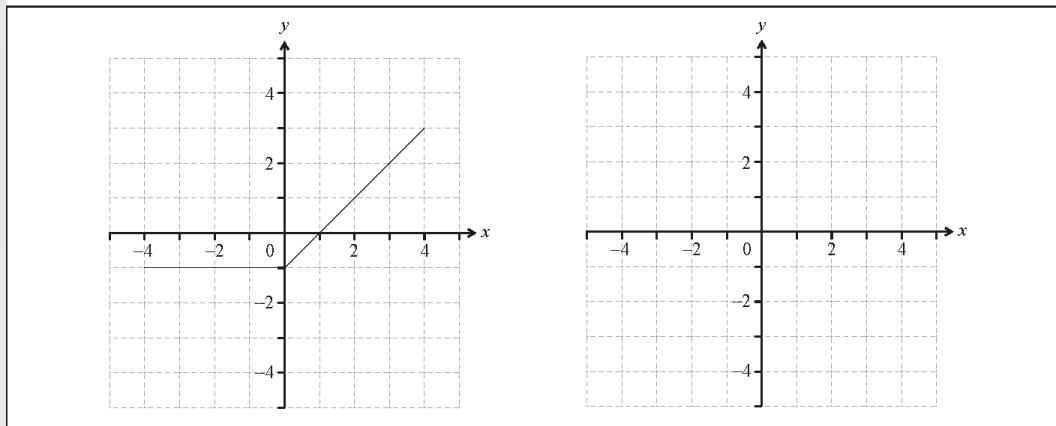
(f) S is rotated through 2π radians about the x -axis. Find the value of the volume generated. [4]

(g) (i) Write down an expression for the constant term in the expansion of $(z + z^{-1})^{2k}$, $k \in \mathbb{Z}^+$.

(ii) Hence determine an expression for $\int_0^{\frac{\pi}{2}} \cos^{2k} \theta \, d\theta$ in terms of k . [3]



The first set of axes below shows the graph of $y = f(x)$ for $-4 \leq x \leq 4$.



(b) On the second set of axes, sketch the graph of $y = g(x)$. [5]

[illegible]

9-(MAT/12_HL_Summer_2014_Q10)-Integration

[Maximum mark: 7]

Use the substitution $x = a \sec \theta$ to show that $\int_{a\sqrt{2}}^{2a} \frac{dx}{x^3 \sqrt{x^2 - a^2}} = \frac{1}{24a^3} (3\sqrt{3} + \pi - 6)$.

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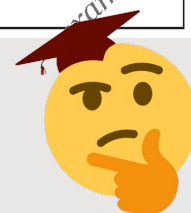
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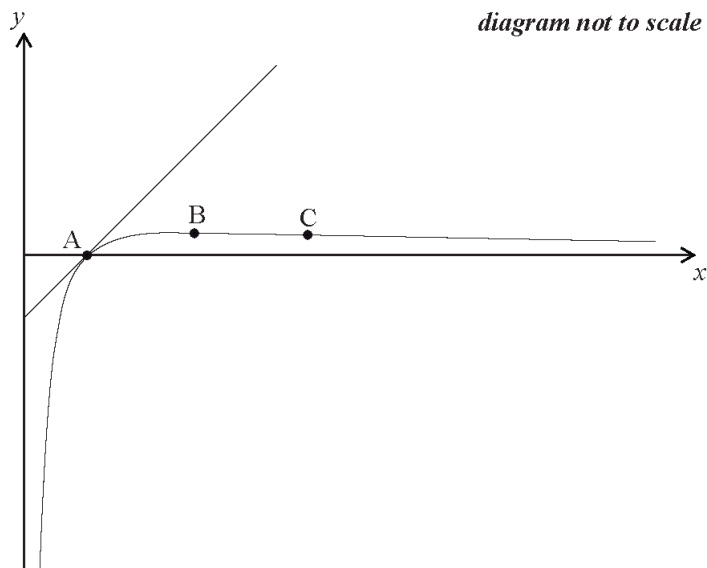
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10-(MAT/11_HL_Summer_2014_Q11)-Differentiation, Integration

Consider the function $f(x) = \frac{\ln x}{x}$, $x > 0$.

The sketch below shows the graph of $y = f(x)$ and its tangent at a point A.



- (a) Show that $f'(x) = \frac{1 - \ln x}{x^2}$. [2]
- (b) Find the coordinates of B, at which the curve reaches its maximum value. [3]
- (c) Find the coordinates of C, the point of inflexion on the curve. [5]

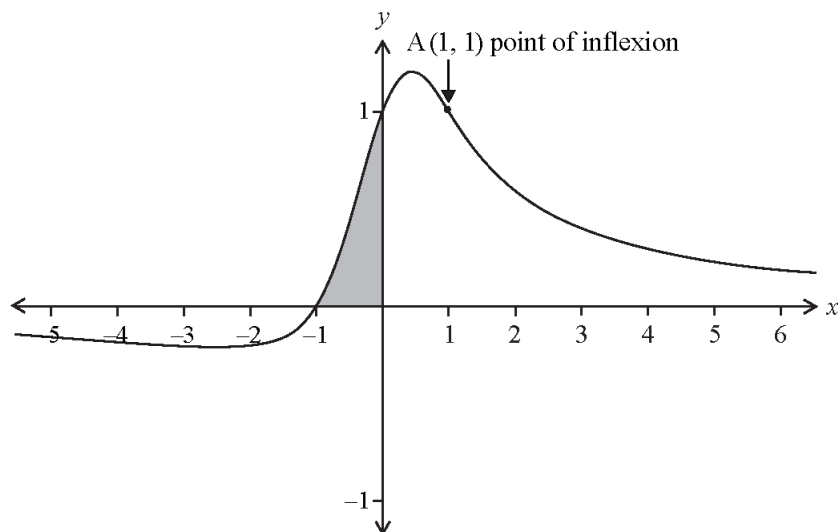
The graph of $y = f(x)$ crosses the x -axis at the point A.

- (d) Find the equation of the tangent to the graph of f at the point A. [4]
- (e) Find the area enclosed by the curve $y = f(x)$, the tangent at A, and the line $x = e$. [7]



11-(MAT/12_HL_Summer_2014_Q13)-Differentiation, Integration

The graph of the function $f(x) = \frac{x+1}{x^2+1}$ is shown below.



- (a) Find $f'(x)$. [2]
- (b) Hence find the x -coordinates of the points where the gradient of the graph of f is zero. [1]
- (c) Find $f''(x)$ expressing your answer in the form $\frac{p(x)}{(x^2+1)^3}$, where $p(x)$ is a polynomial of degree 3. [3]

The point $(1, 1)$ is a point of inflexion. There are two other points of inflexion.

- (d) Find the x -coordinates of the other two points of inflexion. [4]
- (e) Find the area of the shaded region. Express your answer in the form $\frac{\pi}{a} - \ln \sqrt{b}$, where a and b are integers. [6]



12-(MAT/10_HL_Winter_2014_Q6)-Integration

[Maximum mark: 6]

By using the substitution $u = 1 + \sqrt{x}$, find $\int \frac{\sqrt{x}}{1 + \sqrt{x}} dx$.

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13–(MAT/10_HL_Winter_2014_Q11)–Functions-Roots,Differentiation,Integration

The function f is defined as $f(x) = e^{3x+1}$, $x \in \mathbb{R}$.

(a) (i) Find $f^{-1}(x)$.

(ii) State the domain of f^{-1} . [4]

The function g is defined as $g(x) = \ln x$, $x \in \mathbb{R}^+$.

The graph of $y = g(x)$ and the graph of $y = f^{-1}(x)$ intersect at the point P.

(b) Find the coordinates of P. [5]

The graph of $y = g(x)$ intersects the x -axis at the point Q.

(c) Show that the equation of the tangent T to the graph of $y = g(x)$ at the point Q is $y = x - 1$. [3]

A region R is bounded by the graphs of $y = g(x)$, the tangent T and the line $x = e$.

(d) Find the area of the region R . [5]

(e) (i) Show that $g(x) \leq x - 1$, $x \in \mathbb{R}^+$.

(ii) By replacing x with $\frac{1}{x}$ in part (e)(i), show that $\frac{x-1}{x} \leq g(x)$, $x \in \mathbb{R}^+$. [6]



14–(MAT/10_HL_Winter_2014_Q13)–ComplexNumbers,Trigonometry,Integration

- (a) (i) Show that $(1 + i \tan \theta)^n + (1 - i \tan \theta)^n = \frac{2 \cos n\theta}{\cos^n \theta}$, $\cos \theta \neq 0$.
- (ii) Hence verify that $i \tan \frac{3\pi}{8}$ is a root of the equation $(1 + z)^4 + (1 - z)^4 = 0$, $z \in \mathbb{C}$.
- (iii) State another root of the equation $(1 + z)^4 + (1 - z)^4 = 0$, $z \in \mathbb{C}$. [10]
- (b) (i) Use the double angle identity $\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$ to show that $\tan \frac{\pi}{8} = \sqrt{2} - 1$.
- (ii) Show that $\cos 4x = 8 \cos^4 x - 8 \cos^2 x + 1$.
- (iii) Hence find the value of $\int_0^{\frac{\pi}{8}} \frac{2 \cos 4x}{\cos^2 x} dx$. [13]



15-(MAT/11_HL_Summer_2015_Q3)-Integration

(a) Find $\int (1 + \tan^2 x) dx$. [2]

(b) Find $\int \sin^2 x dx$. [3]

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16-(MAT/12_HL_Summer_2015_Q5)-Integration

[Maximum mark: 6]

Show that $\int_1^2 x^3 \ln x \, dx = 4 \ln 2 - \frac{15}{16}$.

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17–(MAT/11_HL_Summer_2015_Q6)–*Functions-Roots, Integration*

A function f is defined by $f(x) = \frac{3x-2}{2x-1}$, $x \in \mathbb{R}$, $x \neq \frac{1}{2}$.

(a) Find an expression for $f^{-1}(x)$. [4]

(b) Given that $f(x)$ can be written in the form $f(x) = A + \frac{B}{2x-1}$, find the values of the constants A and B . [2]

(c) Hence, write down $\int \frac{3x-2}{2x-1} dx$. [1]

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18-(MAT/11_HL_Summer_2015_Q8)-Integration

[Maximum mark: 7]

By using the substitution $u = e^x + 3$, find $\int \frac{e^x}{e^{2x} + 6e^x + 13} dx$.

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19-(MAT/12_HL_Summer_2015_Q8)-Integration

By using the substitution $t = \tan x$, find $\int \frac{dx}{1 + \sin^2 x}$.

Express your answer in the form $m \arctan(n \tan x) + c$, where m, n are constants to be determined.

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20–(MAT/12_HL_Summer_2015_Q11)–Functions-Roots,Differentiation,Integration

Consider the functions $f(x) = \tan x$, $0 \leq x < \frac{\pi}{2}$ and $g(x) = \frac{x+1}{x-1}$, $x \in \mathbb{R}$, $x \neq 1$.

(a) Find an expression for $g \circ f(x)$, stating its domain. [2]

(b) Hence show that $g \circ f(x) = \frac{\sin x + \cos x}{\sin x - \cos x}$. [2]

(c) Let $y = g \circ f(x)$, find an exact value for $\frac{dy}{dx}$ at the point on the graph of $y = g \circ f(x)$ where $x = \frac{\pi}{6}$, expressing your answer in the form $a + b\sqrt{3}$, $a, b \in \mathbb{Z}$. [6]

(d) Show that the area bounded by the graph of $y = g \circ f(x)$, the x -axis and the lines $x = 0$ and $x = \frac{\pi}{6}$ is $\ln(1 + \sqrt{3})$. [6]



21–(MAT/10_HL_Winter_2015_Q2)–Integration

[Maximum mark: 4]

Using integration by parts find $\int x \sin x \, dx$.

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22-(MAT/10_HL_Winter_2015_Q5)-Integration

[Maximum mark: 4]

Use the substitution $u = \ln x$ to find the value of $\int_e^{e^2} \frac{dx}{x \ln x}$.

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23–(MAT/10_HL_Winter_2015_Q12)–Differentiation, Integration

Consider the function defined by $f(x) = x\sqrt{1-x^2}$ on the domain $-1 \leq x \leq 1$.

- (a) Show that f is an odd function. [2]
- (b) Find $f'(x)$. [3]
- (c) Hence find the x -coordinates of any local maximum or minimum points. [3]
- (d) Find the range of f . [3]
- (e) Sketch the graph of $y = f(x)$ indicating clearly the coordinates of the x -intercepts and any local maximum or minimum points. [3]
- (f) Find the area of the region enclosed by the graph of $y = f(x)$ and the x -axis for $x \geq 0$. [4]
- (g) Show that $\int_{-1}^1 |x\sqrt{1-x^2}| dx > \left| \int_{-1}^1 x\sqrt{1-x^2} dx \right|$. [2]



24-(MAT/12_HL_Summer_2016_Q3)-Trigonometry, Integration

(a) Show that $\cot \alpha = \tan\left(\frac{\pi}{2} - \alpha\right)$ for $0 < \alpha < \frac{\pi}{2}$. [1]

(b) Hence find $\int_{\tan \alpha}^{\cot \alpha} \frac{1}{1+x^2} dx$, $0 < \alpha < \frac{\pi}{2}$. [4]

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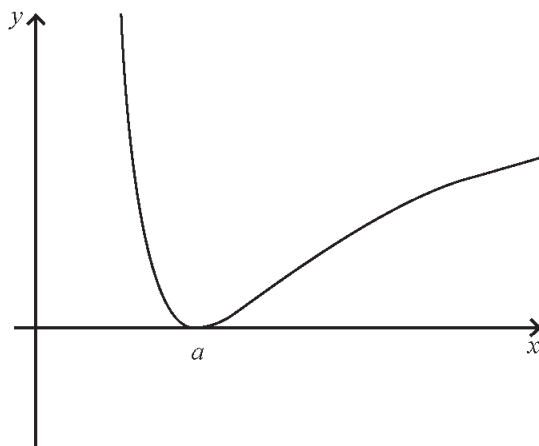
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25-(MAT/11_HL_Summer_2016_Q13)-Integration

The following diagram shows the graph of $y = \frac{(\ln x)^2}{x}$, $x > 0$.



- (a) Given that the curve passes through the point $(a, 0)$, state the value of a . [1]

The region R is enclosed by the curve, the x -axis and the line $x = e$.

- (b) Use the substitution $u = \ln x$ to find the area of the region R . [5]

Let $I_n = \int_1^e \frac{(\ln x)^n}{x^2} dx$, $n \in \mathbb{N}$.

- (c) (i) Find the value of I_0 .
 (ii) Prove that $I_n = -\frac{1}{e} + nI_{n-1}$, $n \in \mathbb{Z}^+$.
 (iii) Hence find the value of I_1 . [7]
- (d) Find the volume of the solid formed when the region R is rotated through 2π about the x -axis. [5]



26-(MAT/12_HL_Summer_2017_Q6)-Integration

(a) Using the substitution $x = \tan \theta$ show that $\int_0^1 \frac{1}{(x^2 + 1)^2} dx = \int_0^{\frac{\pi}{4}} \cos^2 \theta d\theta$. [4]

(b) Hence find the value of $\int_0^1 \frac{1}{(x^2 + 1)^2} dx$. [3]



27–(MAT/11_HL_Summer_2017_Q9)–*Integration*

[Maximum mark: 5]

Find $\int \arcsin x \, dx$.



28—(MAT/12_HL_Summer_2017_Q9)—Functions—Roots, Differentiation, Integration

Consider the function f defined by $f(x) = x^2 - a^2$, $x \in \mathbb{R}$ where a is a positive constant.

- (a) Showing any x and y intercepts, any maximum or minimum points and any asymptotes, sketch the following curves on separate axes.

(i) $y = f(x)$;

(ii) $y = \frac{1}{f(x)}$;

(iii) $y = \left| \frac{1}{f(x)} \right|$. [8]

- (b) Find $\int f(x) \cos x \, dx$. [5]

The function g is defined by $g(x) = x\sqrt{f(x)}$ for $|x| > a$.

- (c) By finding $g'(x)$ explain why g is an increasing function. [4]



29–(MAT/11_HL_Summer_2017_Q11)–Functions-Roots, Graphs, Integration

(a) (i) Express $x^2 + 3x + 2$ in the form $(x + h)^2 + k$.

(ii) Factorize $x^2 + 3x + 2$. [2]

Consider the function $f(x) = \frac{1}{x^2 + 3x + 2}$, $x \in \mathbb{R}$, $x \neq -2$, $x \neq -1$.

(b) Sketch the graph of $f(x)$, indicating on it the equations of the asymptotes, the coordinates of the y -intercept and the local maximum. [5]

(c) Show that $\frac{1}{x+1} - \frac{1}{x+2} = \frac{1}{x^2 + 3x + 2}$. [1]

(d) Hence find the value of p if $\int_0^1 f(x) dx = \ln(p)$. [4]

(e) Sketch the graph of $y = f(|x|)$. [2]

(f) Determine the area of the region enclosed between the graph of $y = f(|x|)$, the x -axis and the lines with equations $x = -1$ and $x = 1$. [3]



30–(MAT/11_HL_Summer_2018_Q4)–Integration

Given that $\int_{-2}^2 f(x)dx = 10$ and $\int_0^2 f(x)dx = 12$, find

(a) $\int_{-2}^0 (f(x) + 2) dx$; [4]

(b) $\int_{-2}^0 f(x+2) dx$. [2]



31–(MAT/12_HL_Summer_2018_Q6)–Differentiation, Integration

Consider the functions f, g , defined for $x \in \mathbb{R}$, given by $f(x) = e^{-x} \sin x$ and $g(x) = e^{-x} \cos x$.

(a) Find

(i) $f'(x)$;

(ii) $g'(x)$.

[3]

(b) Hence, or otherwise, find $\int_0^{\pi} e^{-x} \sin x \, dx$.

[4]



32—(MAT/11_HL_Summer_2018_Q7)—Differentiation, Integration

Let $y = \arccos\left(\frac{x}{2}\right)$.

(a) Find $\frac{dy}{dx}$. [2]

(b) Find $\int_0^1 \arccos\left(\frac{x}{2}\right) dx$. [7]



33–(MAT/12_HL_Summer_2018_Q8)–Integration

(a) Use the substitution $u = x^{\frac{1}{2}}$ to find $\int \frac{dx}{x^{\frac{3}{2}} + x^{\frac{1}{2}}}$. [4]

(b) Hence find the value of $\frac{1}{2} \int_1^9 \frac{dx}{x^{\frac{3}{2}} + x^{\frac{1}{2}}}$, expressing your answer in the form $\arctan q$, where $q \in \mathbb{Q}$. [3]



34–(MAT/12_HL_Summer_2018_Q11)–Exponentials-Logarithms, Integration

- (a) Show that $\log_{r^2} x = \frac{1}{2} \log_r x$ where $r, x \in \mathbb{R}^+$. [2]

It is given that $\log_2 y + \log_4 x + \log_4 2x = 0$.

- (b) Express y in terms of x . Give your answer in the form $y = px^q$, where p, q are constants. [5]

The region R , is bounded by the graph of the function found in part (b), the x -axis, and the lines $x = 1$ and $x = \alpha$ where $\alpha > 1$. The area of R is $\sqrt{2}$.

- (c) Find the value of α . [5]

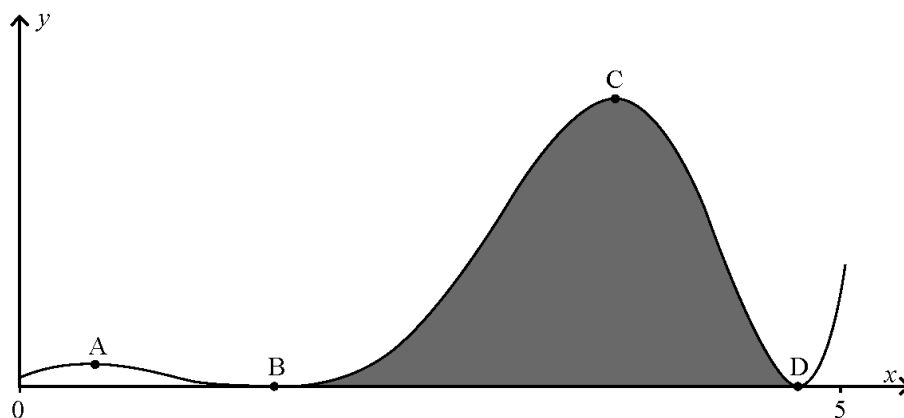


35-(MAT/10_HL_Winter_2018_Q10)-Integration

(a) Use integration by parts to show that $\int e^x \cos 2x dx = \frac{2e^x}{5} \sin 2x + \frac{e^x}{5} \cos 2x + c, c \in \mathbb{R}.$ [5]

(b) Hence, show that $\int e^x \cos^2 x dx = \frac{e^x}{5} \sin 2x + \frac{e^x}{10} \cos 2x + \frac{e^x}{2} + c, c \in \mathbb{R}.$ [3]

The function f is defined by $f(x) = e^x \cos^2 x$, where $0 \leq x \leq 5$. The curve $y = f(x)$ is shown on the following graph which has local maximum points at A and C and touches the x -axis at B and D.



(c) Find the x -coordinates of A and of C, giving your answers in the form $a + \arctan b$, where $a, b \in \mathbb{R}.$ [6]

(d) Find the area enclosed by the curve and the x -axis between B and D, as shaded on the diagram. [5]



36—(MAT/12_HL_Summer_2019_Q4)—Integration

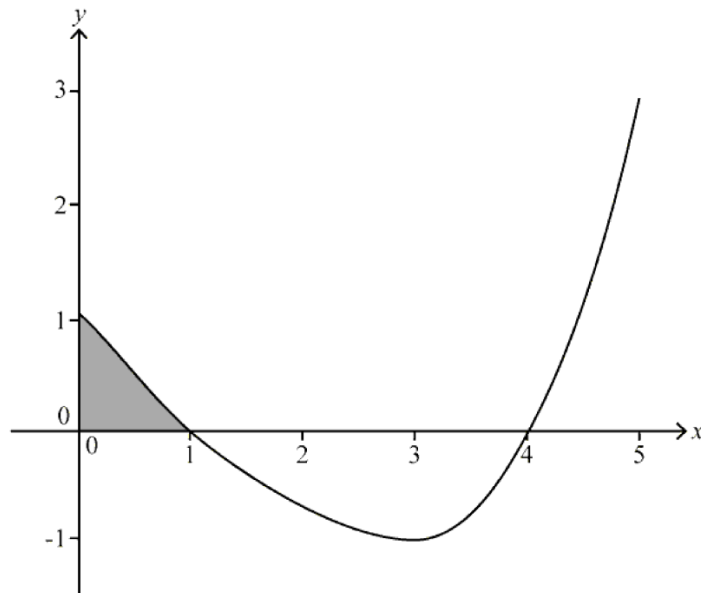
[Maximum mark: 5]

Using the substitution $u = \sin x$, find $\int \frac{\cos^3 x \, dx}{\sqrt{\sin x}}$.



37-(MAT/11_HL_Summer_2019_Q8)-Integration

The graph of $y = f'(x)$, $0 \leq x \leq 5$ is shown in the following diagram. The curve intercepts the x -axis at $(1, 0)$ and $(4, 0)$ and has a local minimum at $(3, -1)$.



- (a) Write down the x -coordinate of the point of inflexion on the graph of $y = f'(x)$. [1]

The shaded area enclosed by the curve $y = f'(x)$, the x -axis and the y -axis is 0.5.
Given that $f(0) = 3$,

- (b) find the value of $f(1)$. [3]

The area enclosed by the curve $y = f'(x)$ and the x -axis between $x = 1$ and $x = 4$ is 2.5.

- (c) Find the value of $f(4)$. [2]

- (d) Sketch the curve $y = f(x)$, $0 \leq x \leq 5$ indicating clearly the coordinates of the maximum and minimum points and any intercepts with the coordinate axes. [3]



38—(MAT/11_HL_Summer_2019_Q9)—Trigonometry, Integration

(a) Show that $(\sin x + \cos x)^2 = 1 + \sin 2x$. [2]

(b) Show that $\sec 2x + \tan 2x = \frac{\cos x + \sin x}{\cos x - \sin x}$. [4]

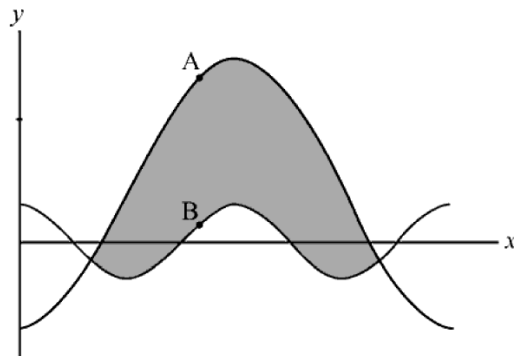
(c) Hence or otherwise find $\int_0^{\frac{\pi}{6}} (\sec 2x + \tan 2x) dx$ in the form $\ln(a + \sqrt{b})$ where $a, b \in \mathbb{Z}$. [9]



39–(MAT/12_HL_Summer_2019_Q9)–Differentiation, Integration

Consider the functions f and g defined on the domain $0 < x < 2\pi$ by $f(x) = 3 \cos 2x$ and $g(x) = 4 - 11 \cos x$.

The following diagram shows the graphs of $y = f(x)$ and $y = g(x)$.



(a) Find the x -coordinates of the points of intersection of the two graphs. [6]

(b) Find the exact area of the shaded region, giving your answer in the form $p\pi + q\sqrt{3}$, where $p, q \in \mathbb{Q}$. [5]

At the points A and B on the diagram, the gradients of the two graphs are equal.

(c) Determine the y -coordinate of A on the graph of g . [6]



40–(MAT/12_HL_Summer_2019_Q10)–Integration, Probability

The random variable X has probability density function f given by

$$f(x) = \begin{cases} k(\pi - \arcsin x) & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}, \text{ where } k \text{ is a positive constant.}$$

(a) State the mode of X . [1]

(b) (i) Find $\int \arcsin x \, dx$.

(ii) Hence show that $k = \frac{2}{2 + \pi}$. [6]

(c) Given that $y = \left(\frac{x^2}{2}\right)\arcsin x - \left(\frac{1}{4}\right)\arcsin x + \left(\frac{x}{4}\right)\sqrt{1-x^2}$, show that

(i) $\frac{dy}{dx} = x \arcsin x$;

(ii) $E(X) = \frac{3\pi}{4(\pi+2)}$. [9]



41–(MAT/10_HL_Winter_2019_Q2)–Integration

[Maximum mark: 6]

Given that $\int_0^{\ln k} e^{2x} dx = 12$, find the value of k .

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42–(MAT/10_HL_Winter_2019_Q7)–Integration

[Maximum mark: 7]

(a) Write $2x - x^2$ in the form $a(x - h)^2 + k$, where $a, h, k \in \mathbb{R}$. [2]

(b) Hence, find the value of $\int_{\frac{1}{2}}^{\frac{3}{2}} \frac{1}{\sqrt{2x - x^2}} dx$. [5]

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43-(MAT/10_HL_Winter_2019_Q10)-Differentiation,Integration

[Maximum mark: 19]

Consider $f(x) = \frac{2x-4}{x^2-1}$, $-1 < x < 1$.

(a) (i) Find $f'(x)$.

(ii) Show that, if $f'(x) = 0$, then $x = 2 - \sqrt{3}$. [5]

(b) For the graph of $y = f(x)$,

(i) find the coordinates of the y -intercept;

(ii) show that there are no x -intercepts;

(iii) sketch the graph, showing clearly any asymptotic behaviour. [5]

(c) Show that $\frac{3}{x+1} - \frac{1}{x-1} = \frac{2x-4}{x^2-1}$. [2]

(d) The area enclosed by the graph of $y = f(x)$ and the line $y = 4$ can be expressed as $\ln v$. Find the value of v . [7]



44–(MAT/10_HL_Winter_2020_Q9)–Integration

[Maximum mark: 8]

By using the substitution $x = \tan u$, find the value of $\int_0^1 \frac{x^2}{(1+x^2)^{3/2}} dx$.



45–(MAT/10_HL_Winter_2020_Q12)–Graphs, Integration

Consider the function defined by $f(x) = \frac{kx-5}{x-k}$, where $x \in \mathbb{R} \setminus \{k\}$ and $k \neq 5$.

- (a) State the equation of the vertical asymptote on the graph of $y = f(x)$. [1]
- (b) State the equation of the horizontal asymptote on the graph of $y = f(x)$. [1]
- (c) Use an algebraic method to determine whether f is a self-inverse function. [4]

Consider the case where $k = 3$.

- (d) Sketch the graph of $y = f(x)$, stating clearly the equations of any asymptotes and the coordinates of any points of intersections with the coordinate axes. [3]
- (e) The region bounded by the x -axis, the curve $y = f(x)$, and the lines $x = 5$ and $x = 7$ is rotated through 2π about the x -axis. Find the volume of the solid generated, giving your answer in the form $\pi(a + b \ln 2)$, where $a, b \in \mathbb{Z}$. [6]



46–(MathAA/12_HL_Summer_2021_Q9)–Integration

[Maximum mark: 7]

By using the substitution $u = \sin x$, find $\int \frac{\sin x \cos x}{\sin^2 x - \sin x - 2} dx$.



47—(MathAA/10_HL_Winter_2021_Q1)—Integration

[Maximum mark: 4]

Given that $\frac{dy}{dx} = \cos\left(x - \frac{\pi}{4}\right)$ and $y = 2$ when $x = \frac{3\pi}{4}$, find y in terms of x .



48—(MathAA/11_HL_Summer_2022_Q1)—Integration

[Maximum mark: 5]

Find the value of $\int_1^9 \left(\frac{3\sqrt{x}-5}{\sqrt{x}} \right) dx$.



49—(MathAA/11_HL_Summer_2022_Q12)—Sequences-Series, Differentiation, Integration

[Maximum mark: 21]

The function f is defined by $f(x) = e^x \sin x$, where $x \in \mathbb{R}$.

(a) Find the Maclaurin series for $f(x)$ up to and including the x^3 term. [4]

(b) Hence, find an approximate value for $\int_0^1 e^{x^2} \sin(x^2) dx$. [4]

The function g is defined by $g(x) = e^x \cos x$, where $x \in \mathbb{R}$.

(c) (i) Show that $g(x)$ satisfies the equation $g''(x) = 2(g'(x) - g(x))$.

(ii) Hence, deduce that $g^{(4)}(x) = 2(g'''(x) - g''(x))$. [5]

(d) Using the result from part (c), find the Maclaurin series for $g(x)$ up to and including the x^4 term. [5]

(e) Hence, or otherwise, determine the value of $\lim_{x \rightarrow 0} \frac{e^x \cos x - 1 - x}{x^3}$. [3]



50—(MathAA/12_HL_Summer_2022_Q7)—Integration

[Maximum mark: 6]

By using the substitution $u = \sec x$ or otherwise, find an expression for $\int_0^{\frac{\pi}{3}} \sec^n x \tan x \, dx$ in terms of n , where n is a non-zero real number.

